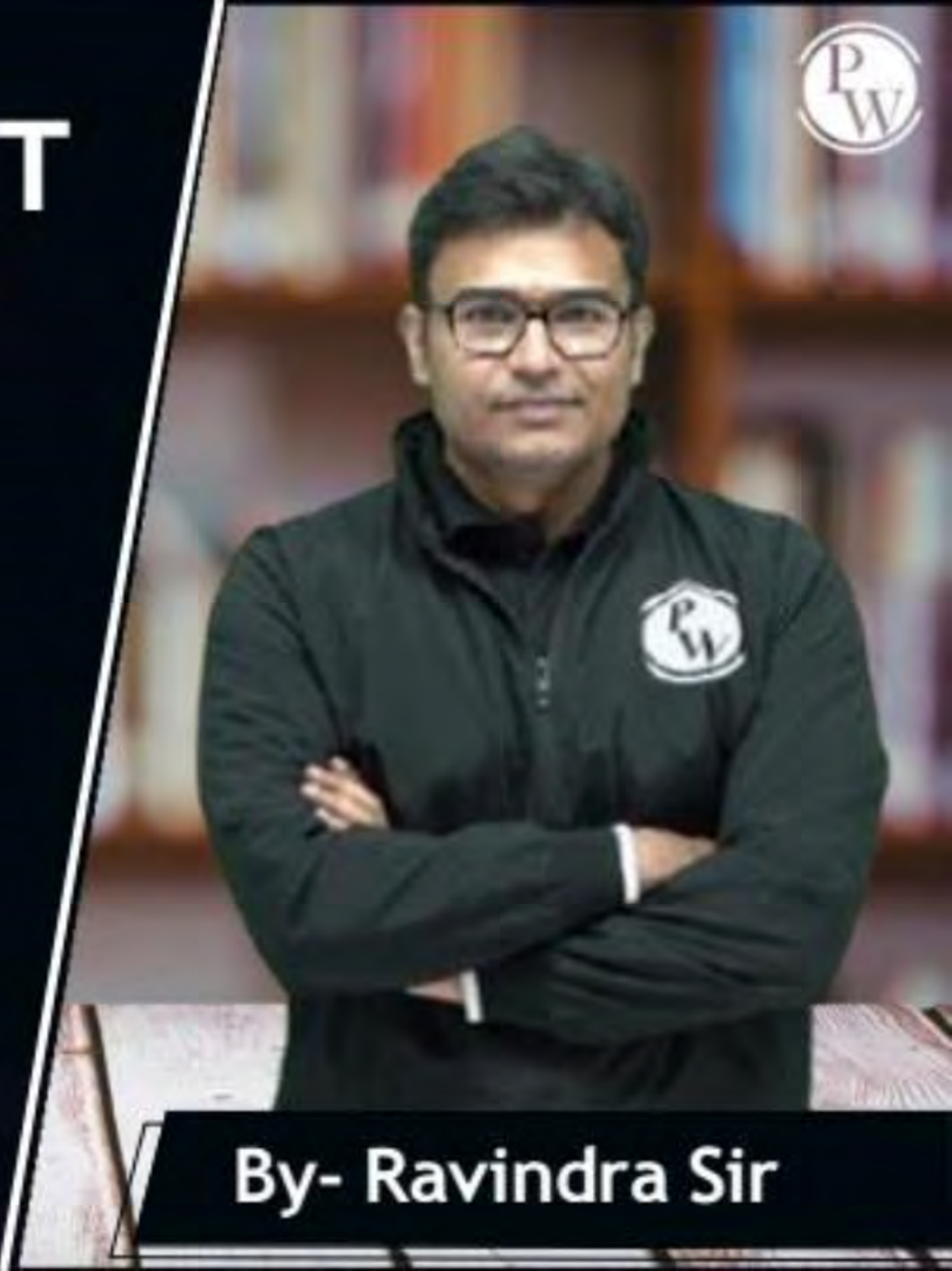
Computer Science & IT

COMPUTER NETWORK
(CN)

TCP and UDP

Lecture No. 01



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

TCP

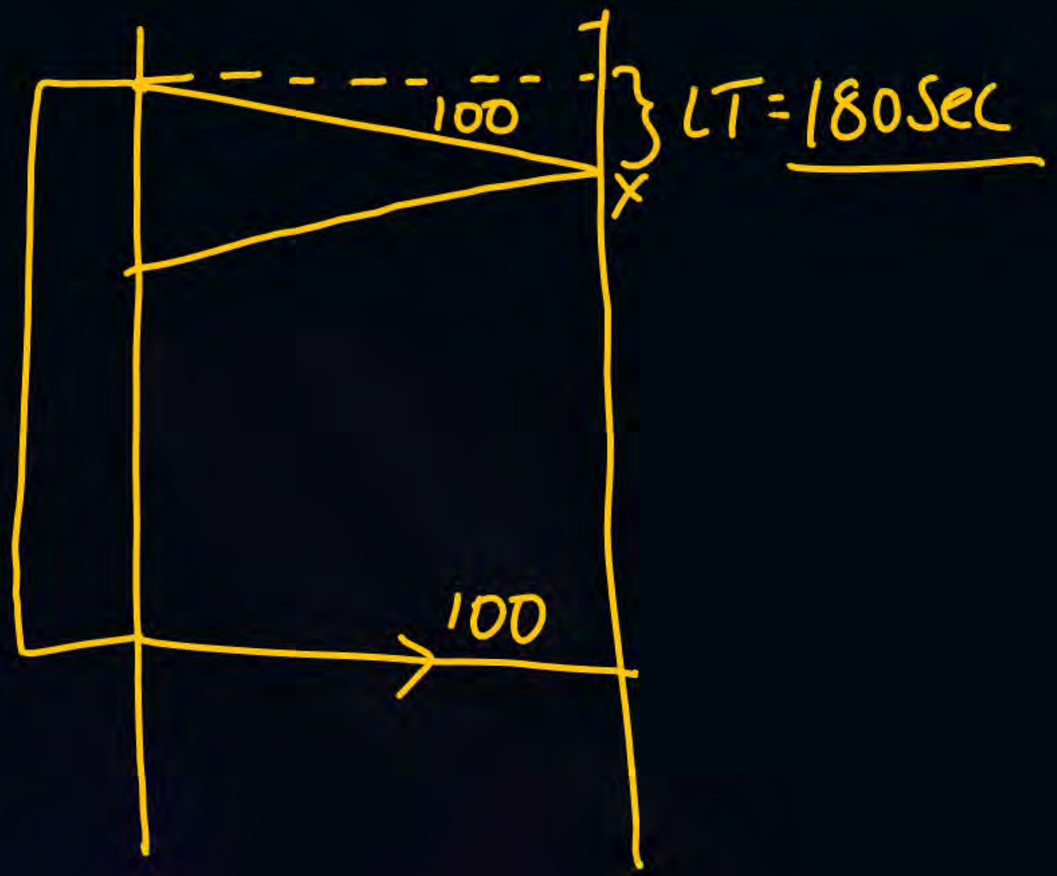
Topics to be Covered



- Topic
- Topic
- Topic

TCP

BW = 1MBPS



WAT: 4296 Sec

$BW = 1\ GBps$

$1\ sec \rightarrow 1\ GB$

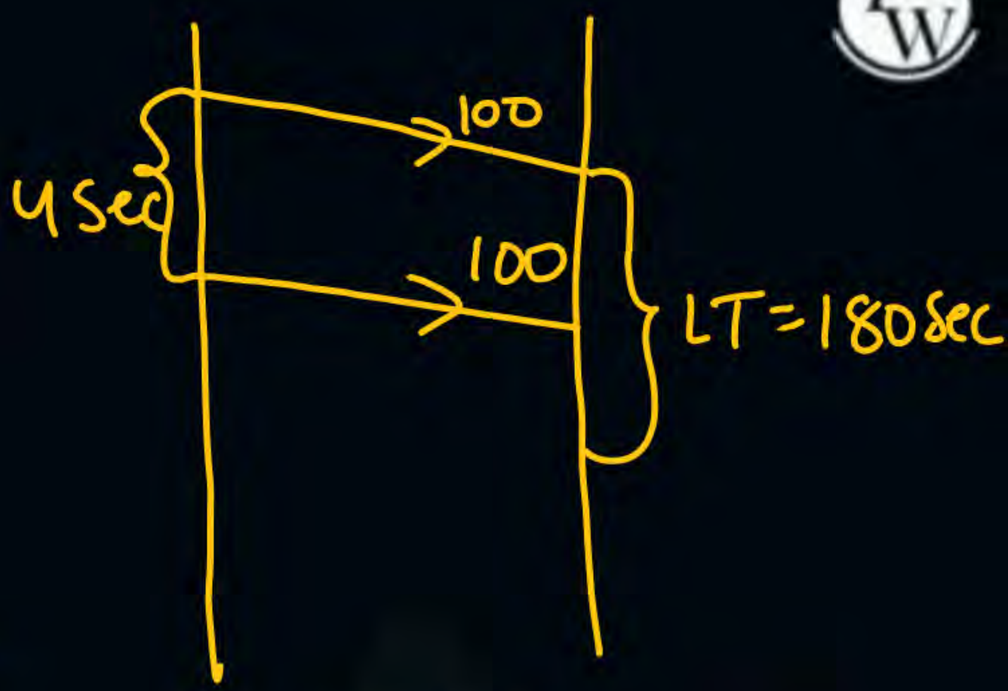
$10^9\ B \rightarrow 1\ sec$

$10^9\ seq \rightarrow 1\ sec$

$1\ seq \rightarrow \frac{1}{10^9}\ sec$

$2^{32}\ seq \rightarrow \frac{2^{32}}{10^9}\ sec$

$WAT = 4.294$



$WAT < LT$

Sol: increase the seq number

LT = 180 Sec, Bw = 1GBps

WAT > LT

WAT > 180 Sec

180 Sec → How many seq numbers we want to avoid WA?

1 Sec — 1 GB

1 Sec — 1G Seq

180 Sec — 180 × 1G Seq

we need extra bits.

we will options

Sequence number field → How many bits?

$\lceil \log 180 \times 10^9 \rceil$ bits → 42 bits → Seq number field

32 — Seq num field

42 → bits

10 extra bit

↓

options

↓

Time stamp

Seq numbers = $\left\lceil \log \left(LT \times \frac{BW}{\text{Bytes}} \right) \right\rceil$

Header length

HLF – 4 bits → max num = 15

Header size → 20B – 60B

Scaling factor $\frac{60}{15} = 4$

Actual header length = 20B

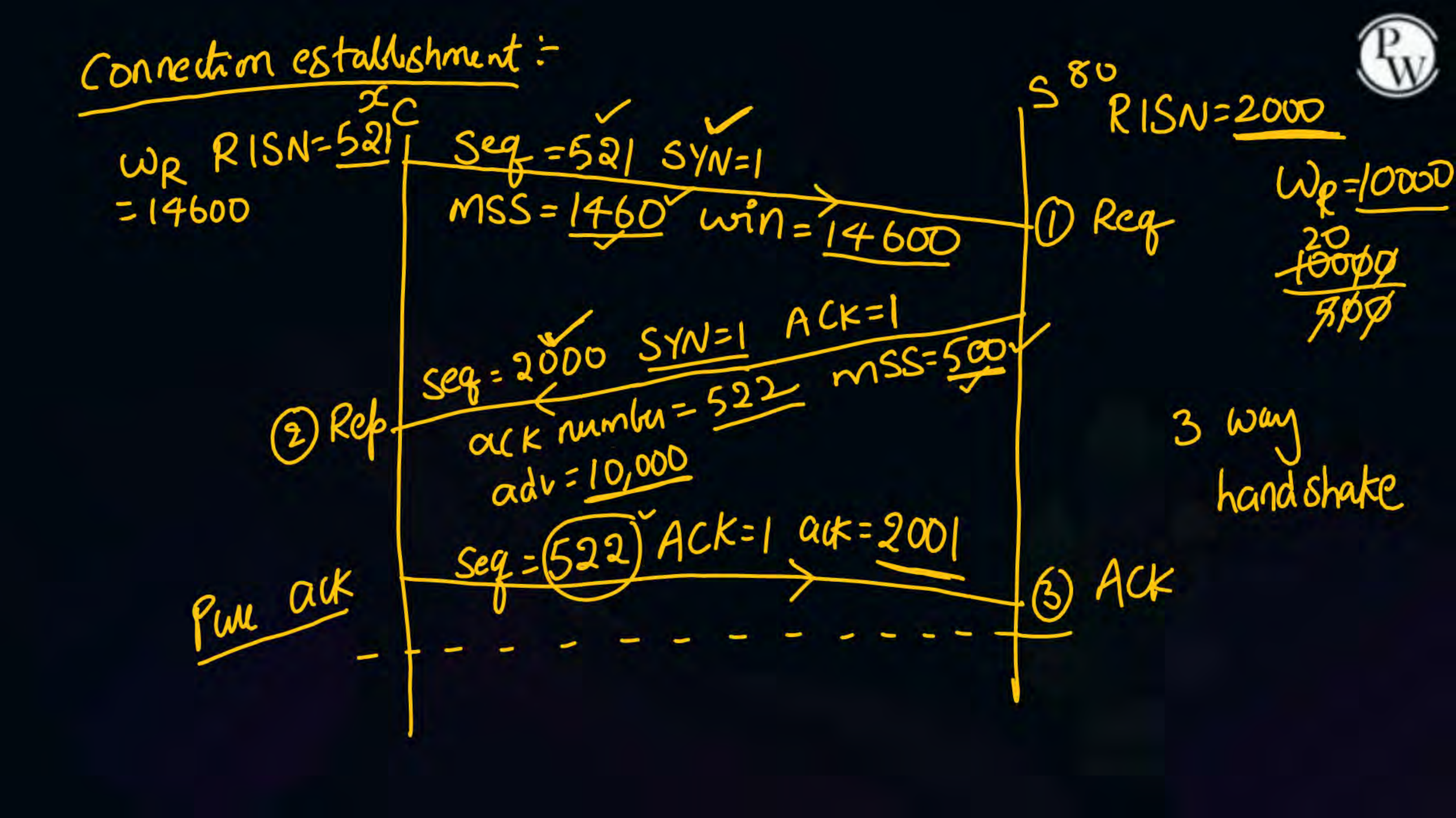
$HLF = \frac{20}{4} = 5$

TCP is a connection oriented protocol :-

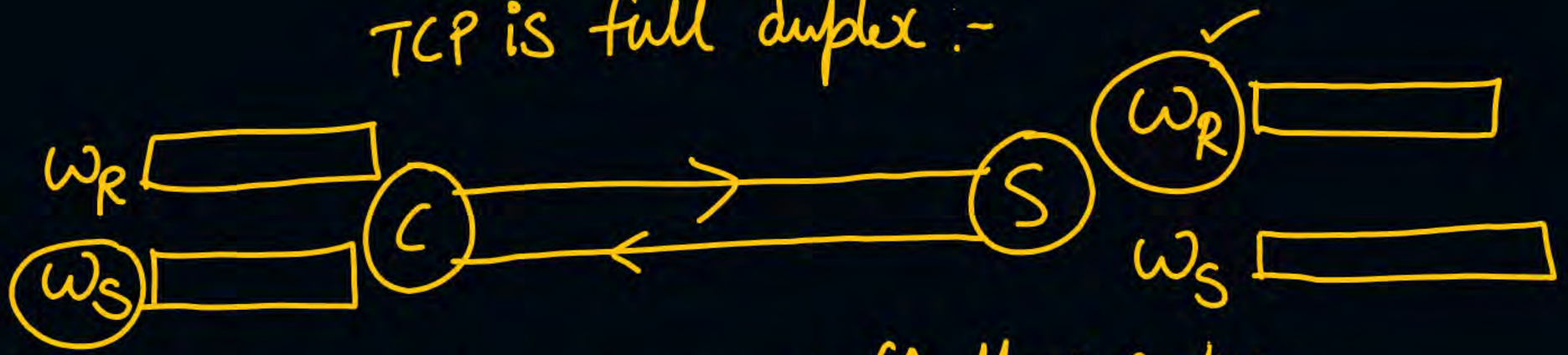
(i) Connection establishment ✓

(ii) Data transfer ✓

(iii) Connection terminate ✓


Connection establishment :-
C
WR RISN=521
=14600
Seq=521 SYN=1
MSS=1460 win=14600
S
RISN=2000
Wp=10000
20
1000
500
① Req
seq=2000 SYN=1 ACK=1
ack number=522 MSS=500
adv=10,000
② Rep
3 way handshake
Seq=522 ACK=1 ack=2001
Pure ack
③ ACK

TCP is full duplex :-



These windows are used for flow contr

$W_S < W_R$

↓

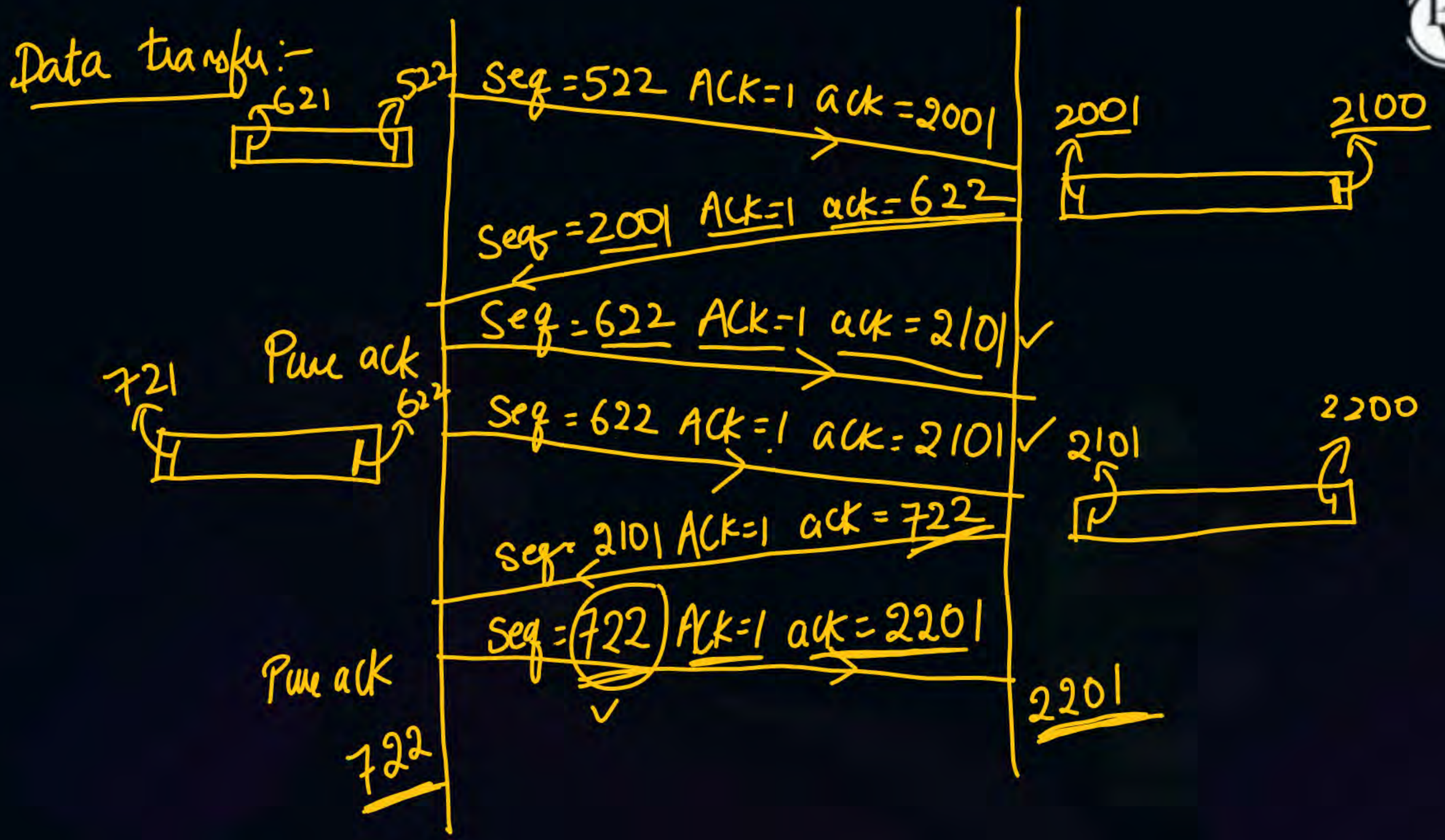
adv window

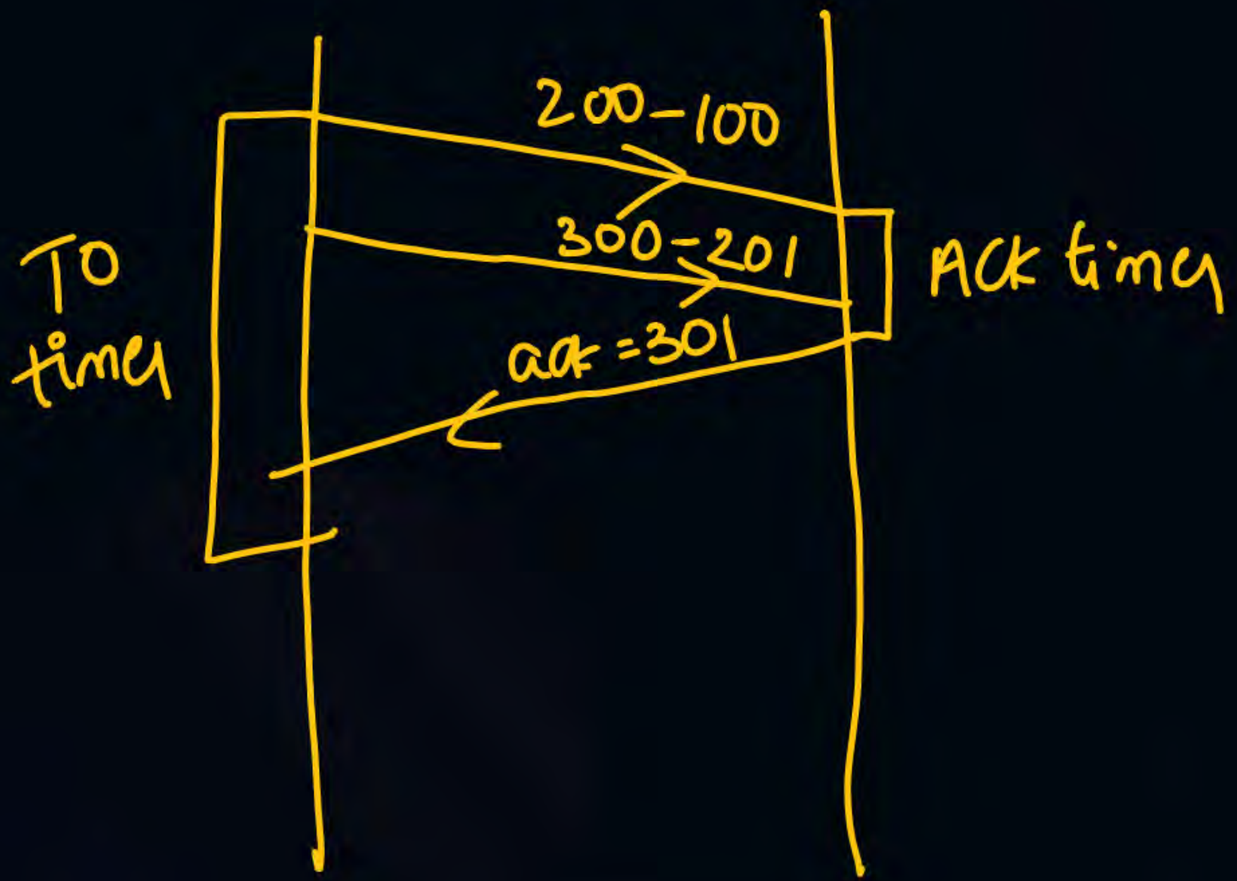
SYN packet will Consume 1 Seq number

Pure ACK packet will not consume any Seq number

Every data byte will Consume 1 Seq number

FIN packet will Consume 1 Seq number


Data transfer:-
621
522
Seq=522 ACK=1 ack=2001
2001
2100
Seq=2001 ACK=1 ack=622
Seq=622 ACK=1 ack=2101
Pure ack
721
622
Seq=622 ACK=1 ack=2101
2101
2200
Seq=2101 ACK=1 ack=722
Seq=722 ACK=1 ack=2201
Pure ack
722
2201

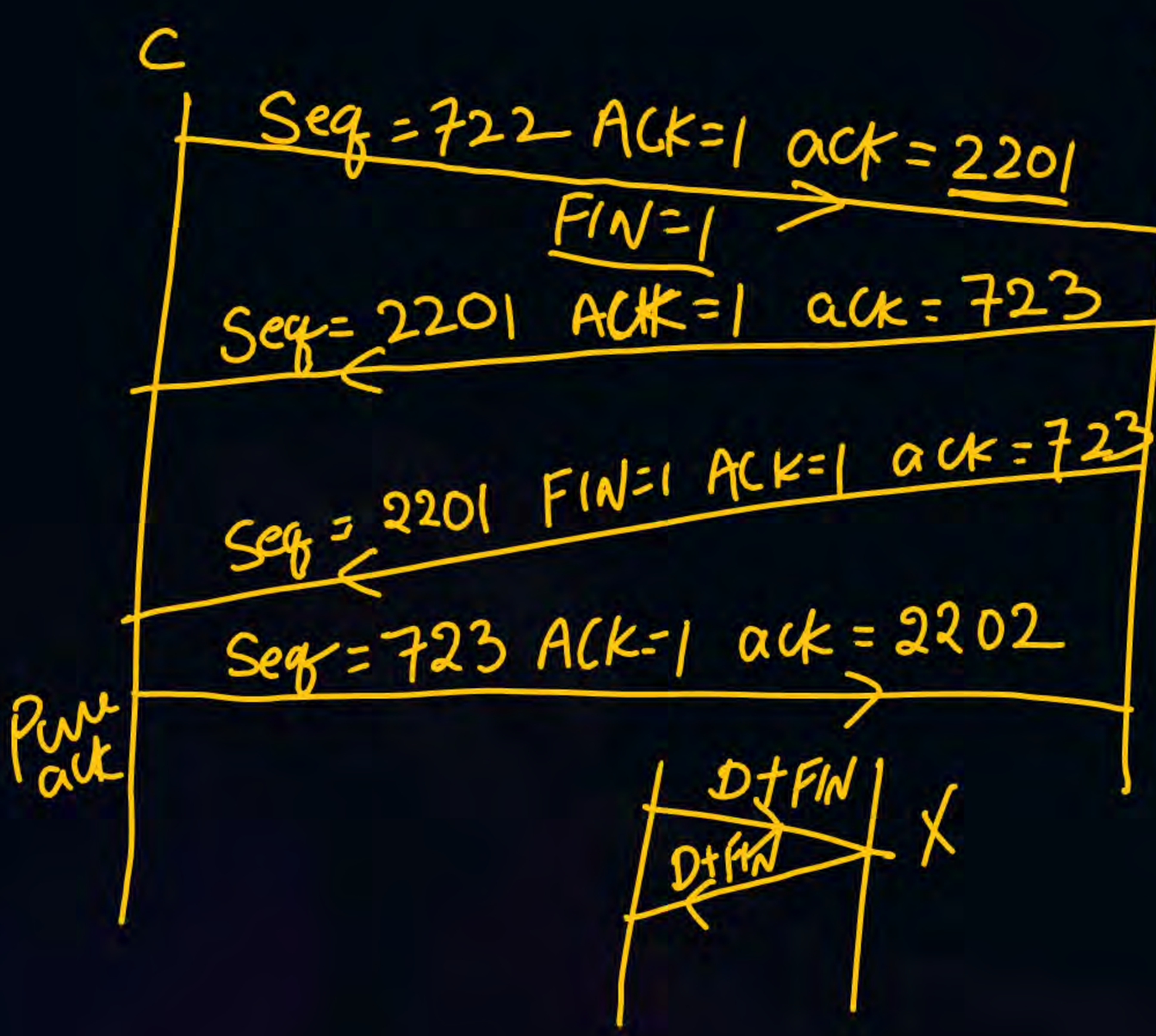

200-100
300-201
ack=301
TO timer
ACK timer

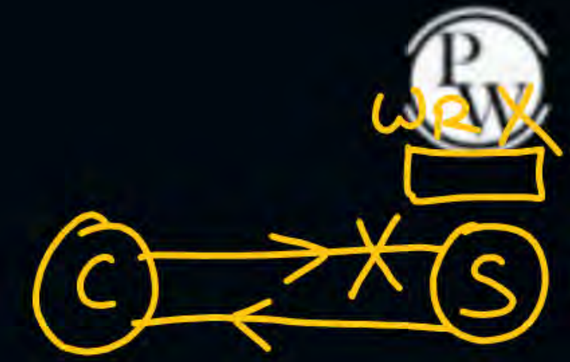
Connection termination :-



Connection establishment → Resources allocate

Connection termination → Resource free





Data: C→S X

Data : S→C ✓

ACK: C→S (Pure) ✓

ACK: S→C (Piggy backing) ✓

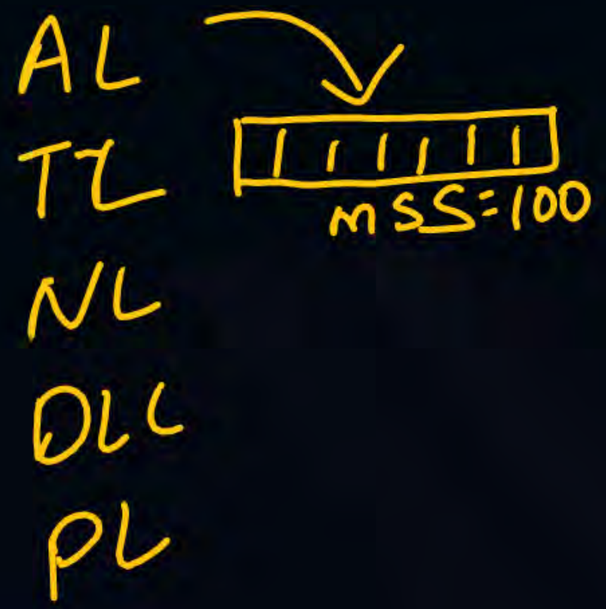


SYN → Synchronize Seq num

ACK → It indicates that packet contains acknowledgment

FIN → Req for connection termination

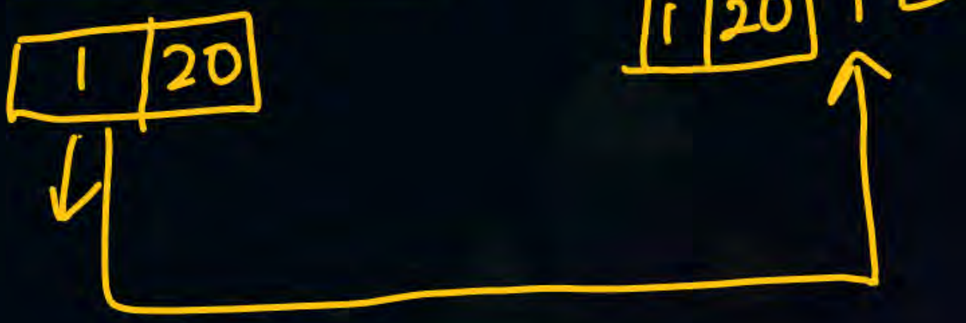
PSH flag (Push):-



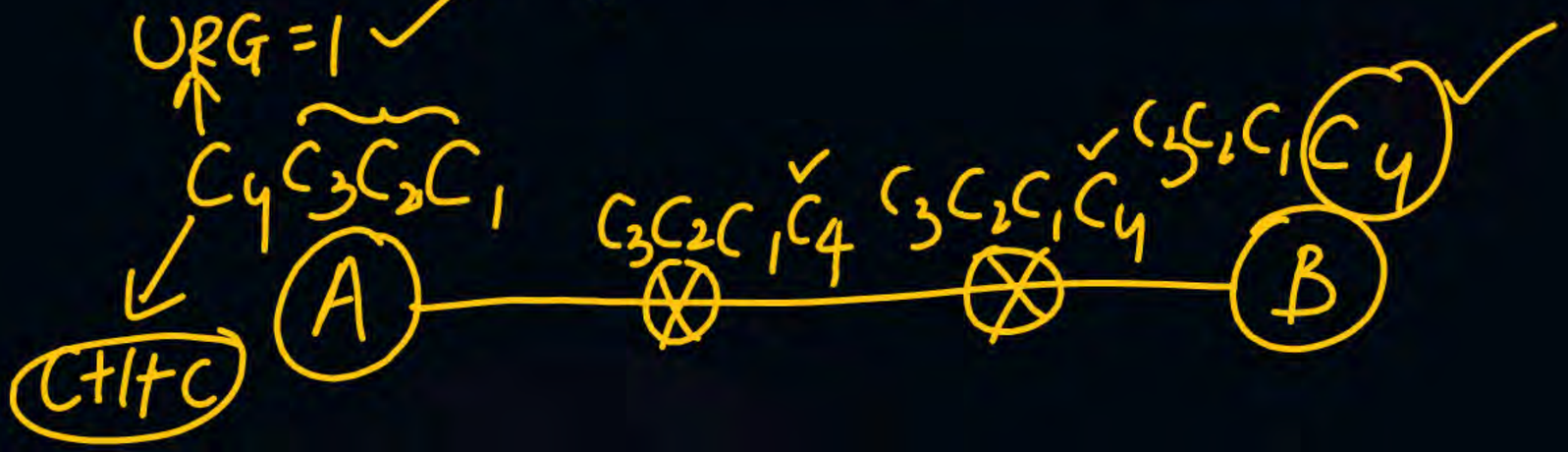
Interactive application

Chat, Telnet, SSH, putty

AL (1Byte, PSH=1)



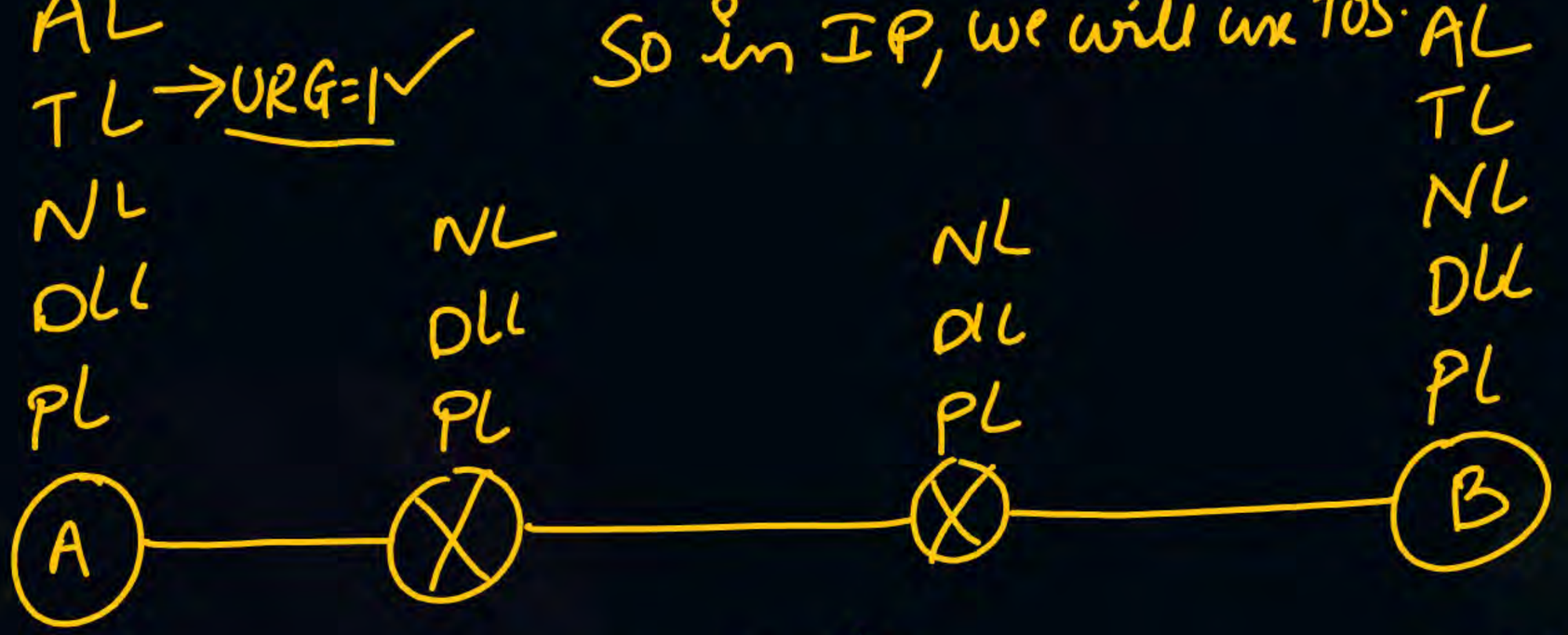
URG flag and urgent pointer :-



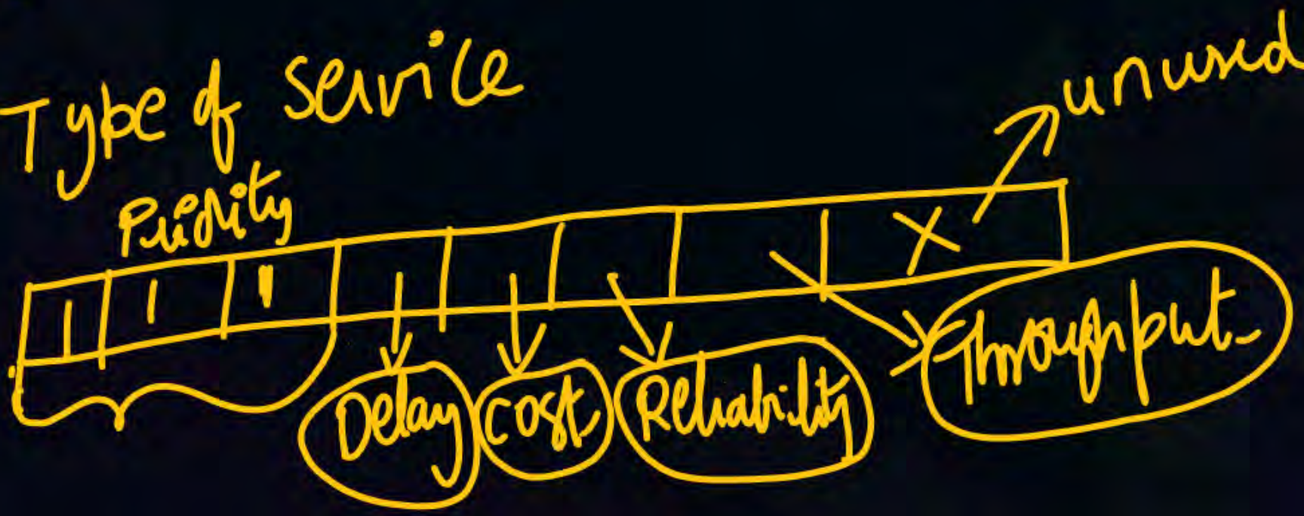
C_4 → has to go first

URG flag is at TL, So routers cannot see URG Flag

So in IP, we will use TOS.



IP → Type of service



When URG=1, the priority in the IP header will be 7

Urgent pointer :-

URG = 1 → It says data is urgent

Urgent pointer → It say how much data is urgent



PYQ:-

Seq = 1000

URG = 1

Urgent pointer = 100

What is the seq number of the last byte that is urgent



UP = 100 → 101

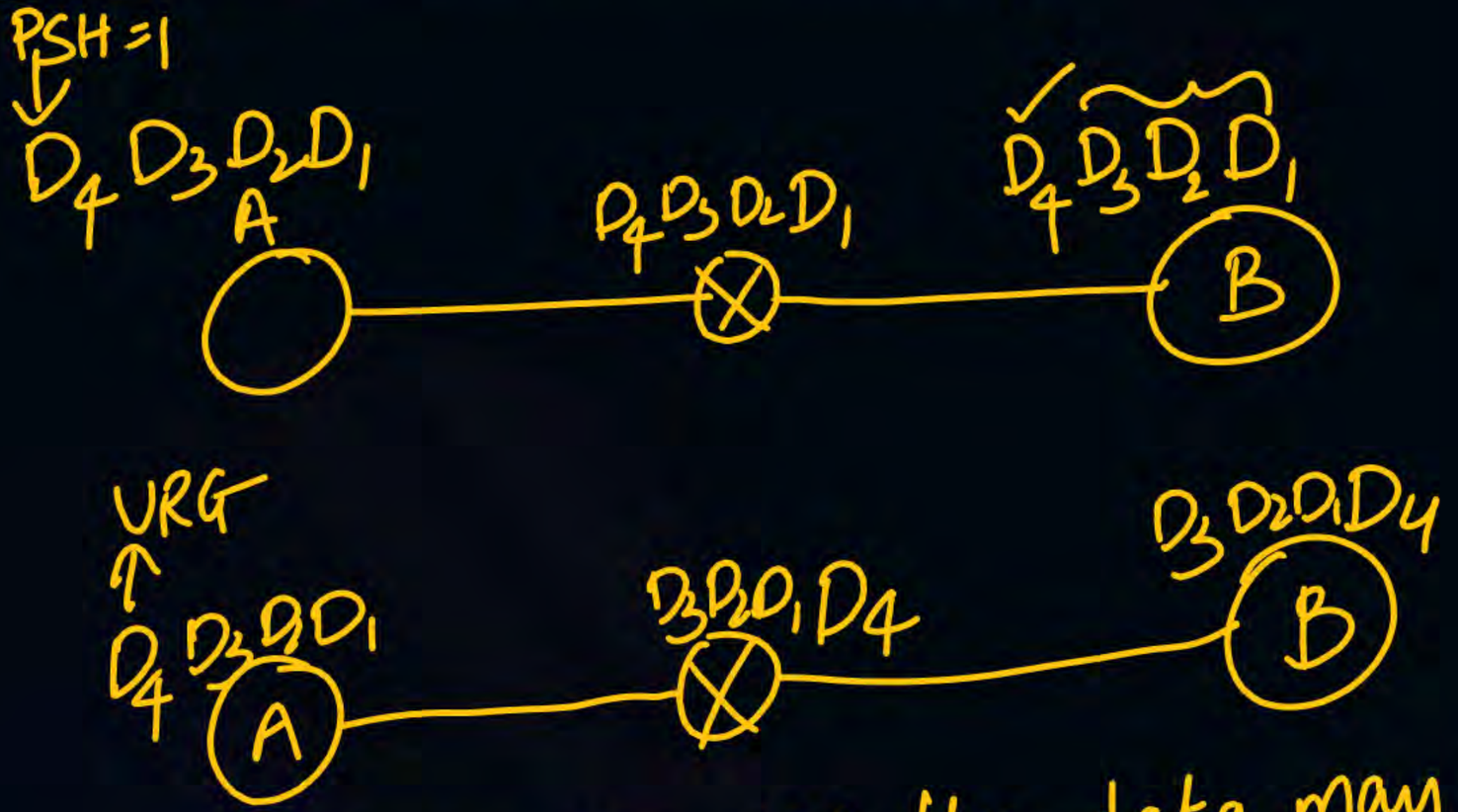
1000 → 1001

1000 + 100 = 1100 → last byte seq number which is urgent

101 ✓

1000 — 1100 → urg

Diff b/w PSH and URG Flag



If we use URG flag, data may go out of order
In case of PSH flag, data will go in order

RST flag (Reset flag) :-



① If B is down in between data transfer and comes up again, it will not remember about connection.

Then B will send RST=1.

RST

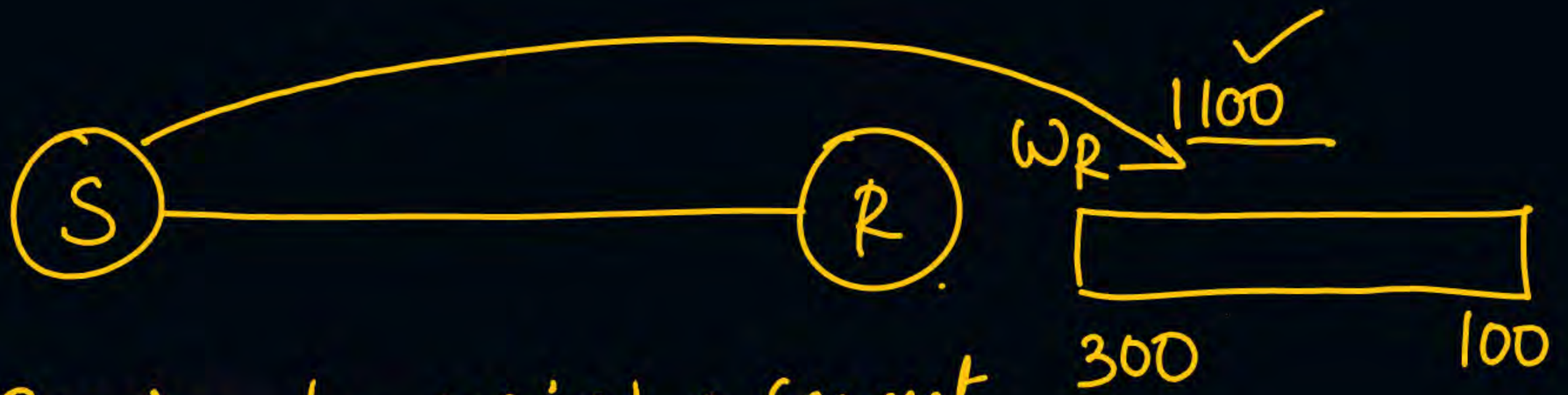
RST=1



There is an attack by C

RST

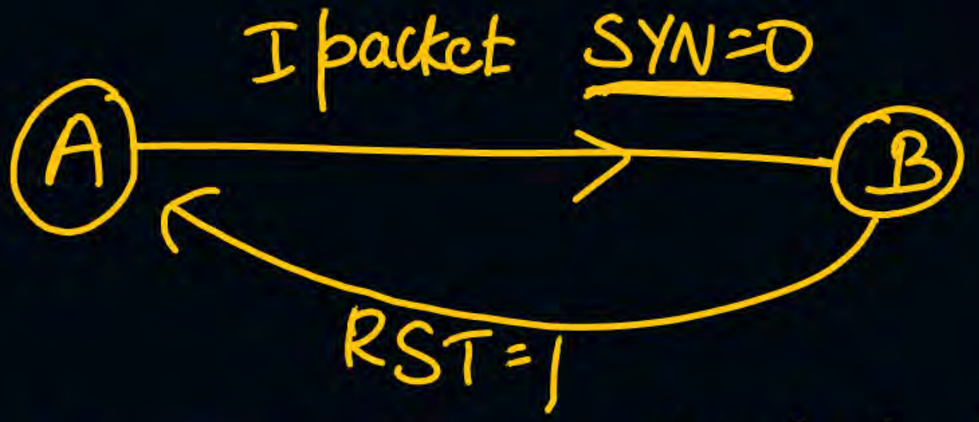
3



Receiver has received a segment which is unexpected

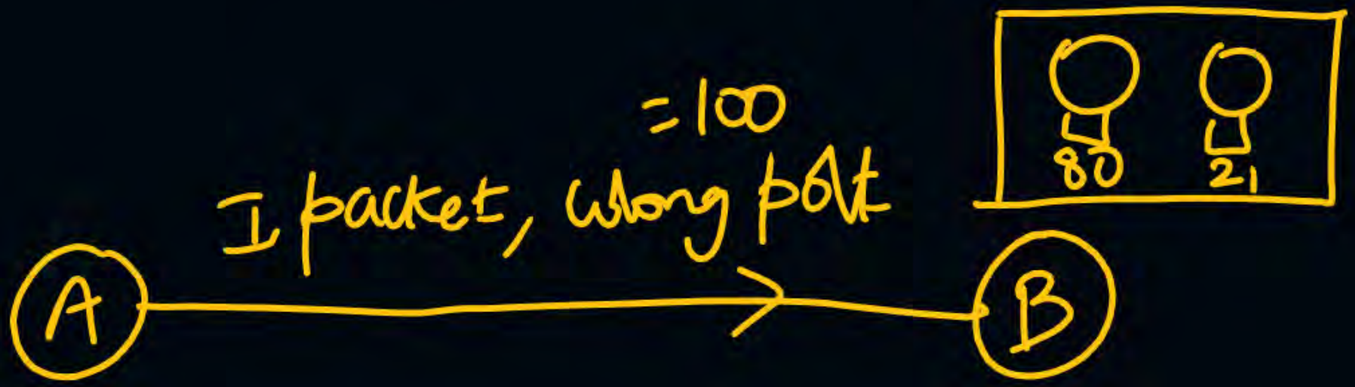
R will send RST = 1

(4)



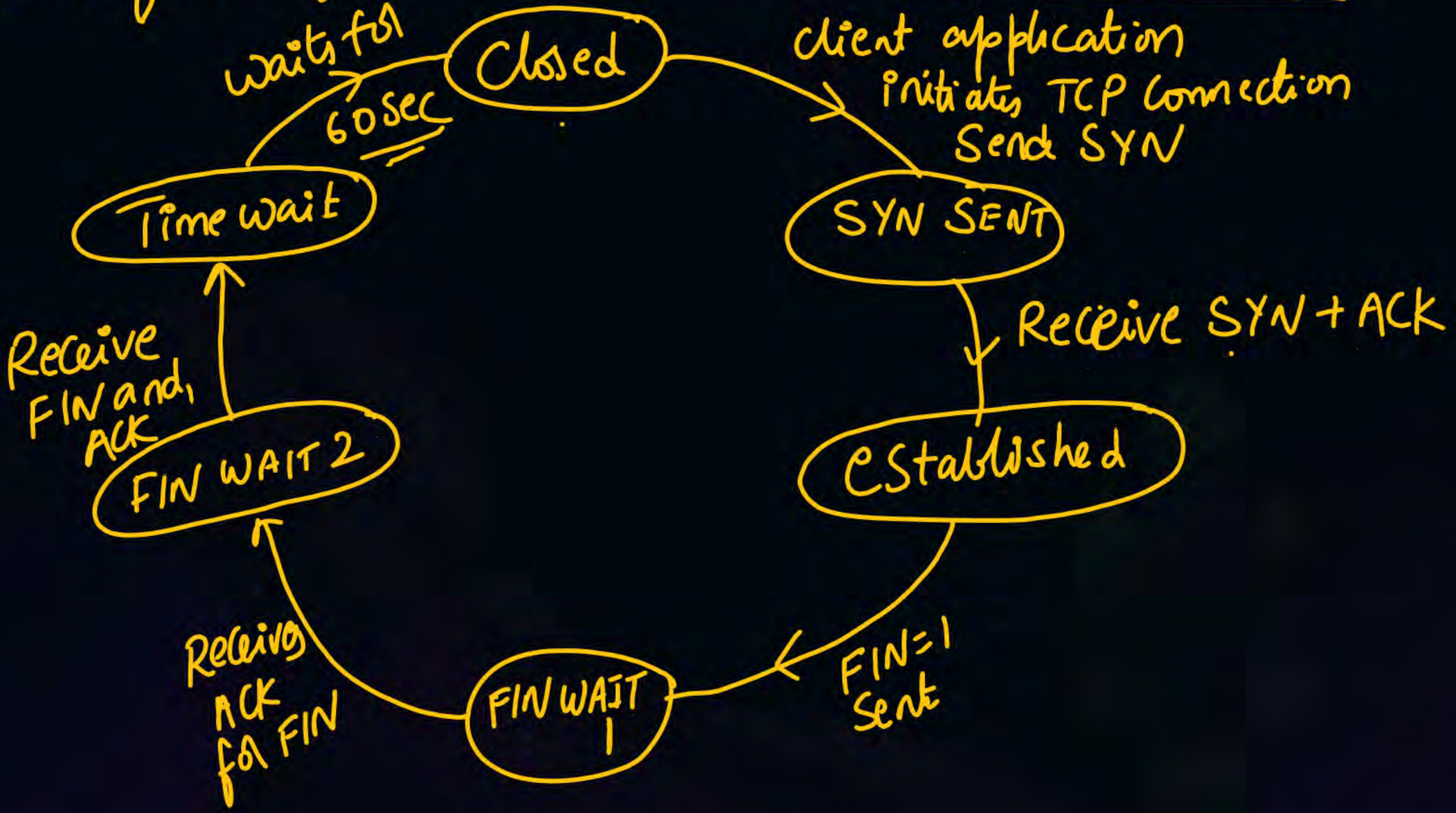
If A sends I packet with SYN=0, then B will Send RST=1

(5)



when a host sends a message on a wrong port number then it will receive RST = 1

The Sequence of TCP states visited by Client TCP



THANK - YOU